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Roll No. | | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 09

B.Tech. (CE) (Sem.–6)

GROUND IMPROVEMENT TECHNIQUES

Subject Code : PECE-602B-18
M.Code : 79395
Date of Examination : 16-12-2022

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION-B contains FIVE questions carrying FIVE marks each and students have to attempt any FOUR questions.**
3. **SECTION-C contains THREE questions carrying TEN marks each and students have to attempt any TWO questions.**

SECTION-A

1. Write briefly :

a) What precautions should be taken while using Black cotton soil for highways?

b) What are the conditions which make ground improvement necessary?

c) What is the principle of pre-compaction?

d) What are sand and band drains?

e) What are the chemicals used for grouting?

f) List applications of soil nailing.

g) What are soil dowels?

h) What do you mean by improvement ratio?

i) What are the various stability checks applied while designing geo-synthetic wall?

j) When are stone columns preferred?

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SECTION-B

2. Write steps of improvement of CBR by various ground improvement techniques.

3. Discuss dynamic consolidation technique. Give its disadvantages. Why is dynamic consolidation effective in soft clays? Give reasons.

4. Explain grouting technology of ground improvement. Discuss Compaction grouting.

5. Describe the steps of stabilization of slope by external reinforcement.

6. Write steps for design of retaining wall with reinforced backfill.

SECTION-C

7. **Write short notes on:**

   a) Improvement of black cotton soil

   b) Improvement of lateritic soil

   c) Ground improvement of cohesive soils

   d) Ground improvement of granular soils.

8. What are the various vibro compaction methods used for densification? Explain in detail.

9. Describe use of geosynthetics in road design.

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